Task 3a: A silent bug

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The database's leads_source_check permits only seven tool names. It omits battle_card_generator and case_study_generator, so these inserts fail with PostgreSQL check-constraint error 23514. postToZohoForm() runs first without being awaited, independently of the database write. The Supabase failure is only logged; returning data?.id gives the caller undefined, so the later session save cannot link to a real lead while the interface may still show success.

I would add the two missing source values in a versioned migration, keeping the existing seven. The frontend, API validation and database must share the same source contract, including linkedin_hook_grader when the tenth tool ships. I would validate input, save the lead and its tool session together in a database transaction, and fail visibly unless both commit. That transaction would also create a durable CRM-sync job. A worker would await the Zoho response, record its outcome and retry temporary failures with an idempotency key. A database transaction cannot make an external Zoho request atomic.

To prevent recurrence, I would test every allowed source, a rejected unknown source, database failures and CRM retries; alert on persistent insert/sync errors; and reconcile completed tool runs against Supabase sessions and CRM records by source and session ID. I would recover missing sessions from retained request/execution logs where available, preserving original IDs and timestamps, and label unrecoverable inputs/outputs instead of inventing them. I would not blindly replay CRM submissions and create duplicate contacts.

Analysis of the supplied snippet. No production schema changes were executed.